Grade: Check Plus

Assignment 2
Expected Values and Options

Price x	Prob. f(x)	x f(x)	$(x-25.41)^2 f(x)$	✓
20	0.05	1	1.46	0
21	0.06	1.26	1.17	0
22	0.07	1.54	0.81	0
23	0.08	1.84	0.46	0
24	0.09	2.16	0.18	0
25	0.12	3	0.02	0
26	0.15	3.9	0.05	1
27	0.13	3.51	0.33	2
28	0.10	2.8	0.67	3
29	0.10	2.9	1.29	4
30	0.05	1.5	1.05	5
	1.00	25.41	7.49	

1. Compute the mean and standard deviation of the stock price. $\boxed{\mu = 25.41}$

$20 \times 0.05 = 1$
$21 \times 0.06 = 1.26$
$22 \times 0.07 = 1.54$
$23 \times 0.08 = 1.84$
$24 \times 0.09 = 2.16$
$25 \times 0.12 = 3$
$26 \times 0.15 = 3.9$
$27 \times 0.13 = 3.51$
$28 \times 0.10 = 2.8$
$29 \times 0.10 = 2.9$
$30 \times 0.05 = 1.5$

$\mu = 1 + 1.26 + 1.54 + 1.84 + 2.16 + 3 + 3.20 + 3.51 + 2.8 +$
$2.9 + 1.5 = \boxed{25.41}$

$$\sigma^2 = \sum_{all\,x} (x - 25.41)^2 f(x) = 7.49$$

$(20 - 25.41)^2 \times 0.05 = 1.46$
$(21 - 25.41)^2 \times 0.06 = 1.17$
$(22 - 25.41)^2 \times 0.07 = 0.81$
$(23 - 25.41)^2 \times 0.08 = 0.46$
$(24 - 25.41)^2 \times 0.09 = 0.18$
$(25 - 25.41)^2 \times 0.12 = 0.02$
$(26 - 25.41)^2 \times 0.15 = 0.05$
$(27 - 25.41)^2 \times 0.13 = 0.33$
$(28 - 25.41)^2 \times 0.10 = 0.67$
$(29 - 25.41)^2 \times 0.10 = 1.29$
$(30 - 25.41)^2 \times 0.05 = 1.05$
7.49

✓ Q1

$\sigma = \sqrt{7.49} = \boxed{2.74}$

2a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?

$P(25.41 - 2.74 \leq X \leq 25.41 + 2.74)$
$P(22.67 \leq X \leq 28.15)$
$P(23 \leq X \leq 28)$

$P(23) + P(24) + P(25) + P(26) + P(27) + P(28)$ ✓ Q2a

$0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10 = \boxed{0.67 \text{ or } 67\%}$

2b. What is the probability that the stock price will fall within 2 standard deviations of the mean a year from now?

$P(25.41 - 2 \times 2.74 \leq X \leq 25.41 + 2 \times 2.74)$

$P(19.93 \leq X \leq 30.89)$

$P(20 \leq X \leq 30)$

$P(20) + P(21) + P(22) + P(23) + P(24) + P(25) + P(26) + P(27) + P(28) + P(29) + P(30)$

$0.05 + 0.06 + 0.07 + 0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10 + 0.10 + 0.05$ ✓ Q2b

$= \boxed{1.00 \text{ or } 100\%}$

3. Derive the distribution of the option value, ignoring interest and commissions.

x	$f(x)$	v
20	0.05	0
21	0.06	0
22	0.07	0
23	0.08	0
24	0.09	0
25	0.12	0
26	0.15	1
27	0.13	2
28	0.10	3
29	0.10	4
30	0.05	5

v	$g(v) = P(V=v)$
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05
	1.00

✓ Q3

4. What is the probability that the option will have a positive value?

$P(X=26) = 0.15$
$P(X=27) = 0.13$
$P(X=28) = 0.10$
$P(X=29) = 0.10$
$P(X=30) = 0.05$

0.53 ✓ Q4

There is a 53% probability that the option will have a positive value.

5. Compute the mean and standard deviation of the value of the option.

$1(0.15) = 0.1$
$2(0.13) = 0.26$
$3(0.10) = 0.30$
$4(0.10) = 0.40$
$5(0.05) = 0.25$

$\boxed{\mu = 1.36}$

$Var(V) = \Sigma(v-\mu)^2 P(v$

$0 = (0-1.36)^2 \times 0.47 = 0.87$
$1 = (1-1.36)^2 \times 0.15 = 0.02$
$2 = (2-1.36)^2 \times 0.13 = 0.05$
$3 = (3-1.36)^2 \times 0.10 = 0.27$
$4 = (4-1.36)^2 \times 0.10 = 0.70$
$5 = (5-1.36)^2 \times 0.05 = 0.66$

$Var = 2.57$

$\sigma_V = \sqrt{2.57}$ ✓ Q5

$\boxed{\sigma_V = 1.60}$